Venkata Lakshmi Parimala Pasupuleti

Staff AI Platform Engineer · Python · AWS · PostgreSQL · LLM Evaluation · Event-Driven Architecture · GTM Intelligence

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Virginia, USA — Open to Austin, TX

PROFESSIONAL SUMMARY

Staff-scope engineer with **almost 10 years shipping production systems** (Python, AWS, PostgreSQL, event-driven architectures, 100K+ MAU) and hands-on AI platform expertise: LLM evaluation frameworks, AI governance, and multi-agent observability. Built **AgentProbe** - 5-metric LLM-as-judge eval engine with CI/CD regression gates and real-time drift tracking. At Entrata, owned 20+ production modules, led microservices redesign (50% scalability gain), and built CRM data pipelines to commercial systems via Salesforce, Clay, and n8n. M.S. Data Analytics Engineering, GMU (GPA 3.87).

CORE COMPETENCIES

AI Governance & LLM Evaluation | Event-Driven Architecture | Enterprise Platform Engineering | Observability & Reliability | GTM & CRM Data Pipelines | Python · AWS · PostgreSQL | Multi-Agent Orchestration (LangGraph) | Technical Mentorship

WORK EXPERIENCE

Entrata — PropTech SaaS (100K+ Monthly Active Users)

Jan 2017 – Aug 2024

Senior Software Engineer → Associate Software Engineer → Trainee — Full-Stack Production Engineering

- Owned architecture of **20+ full-stack production modules** (Python, PostgreSQL, RabbitMQ event-driven architecture) across 120+ Agile sprints for 100K+ MAU — 99.5% uptime; mentored 5+ junior engineers with structured code reviews and debugging sessions.
- Led **microservices architecture redesign** (Docker) — 50% scalability improvement, 30% deployment time reduction; converted 20% of modules to idempotent service-based architecture.
- **Built CRM data pipelines** to commercial GTM systems — Salesforce integrations, reporting automations, real-time data sync via Clay and n8n; automated account data workflows replacing manual GTM operations.
- Shipped **first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) with output validation, quality gates, and compliance controls; achieved 100% PHI compliance on AWS S3 migration (zero incidents); optimized 200+ SQL queries (30% gain).

George Mason University — College of Engineering and Computing

Aug 2025 – Present

Graduate Teaching Assistant — Instructional Leadership · 600+ Students

- Lead hands-on lab sessions and office hours for **600+ students across multiple sections**; coordinate content delivery and pacing with lead instructor — cross-functional stakeholder management at scale.

George Mason University — Research

Aug 2024 – Aug 2025

Research Associate — Data Pipelines · Statistical Modeling · Executive Communication

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries — $R^2 = 0.977$ accuracy; **improved research comprehension 40%** across two faculty cohorts via Tableau and ggplot2 executive dashboards; surfaced anomalies manual review missed by applying Isolation Forest across 3 research pipelines.

Magizhs Tech Solution — AI Intern

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Mission Planning · LangGraph · RAG

- Built an agentic mission-planning system (LangGraph stateful graph, RAG over FAA Part 107 corpus, validation guardrails) with deterministic compliance routing — non-compliant missions blocked before export with grounded rejection reasons.

PROJECTS

AgentProbe — Agentic AI Testing Toolkit

Python · LangGraph · FastAPI · ChromaDB · React · Docker · GitHub Actions

- Built **AI governance infrastructure**: 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD integration — every commit auto-evaluated, regression gates blocking failing runs before production.
- Delivered **real-time observability** across concurrent agent test suites — per-run metrics, regression drift, retrieval quality, and latency — via React dashboard backed by ChromaDB RAG pipeline and 15-endpoint FastAPI backend.
- Multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) across cloud and on-premises inference — cost management and provider failover built in. Live: github.com/venkatalakshmiparimala/AgentProbe

Leasel — AI Compliance System

Python · LangGraph · SQLite · LLM-as-Judge · Deterministic Rules Engine

- 93.6% precision and 87.9% F1 across 1,240 residential lease PDFs — agentic RAG pipeline with deterministic rules engine keeping all compliance determinations fully auditable; pass@1 81.7%, pass@3 88.3% via multi-trial sampling and regression gate.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87
Fairfax, VA

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

SKILLS

Languages & Platforms: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

AI & LLM Eval: LangChain, LangGraph, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, AI Governance, Observability, ChromaDB, RAG Pipelines, Prompt Engineering, OpenAI API, Anthropic API, FastAPI

Data & ML: PyTorch, Scikit-learn, Isolation Forest, Feature Engineering, Statistical Modeling, Pandas, NumPy, Tableau, Power BI

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Idempotent APIs, Event-Driven Architecture (RabbitMQ), Microservices, PostgreSQL, MySQL, Oracle, Libsodium, Agile/Scrum

GTM & CRM: Salesforce integrations, reporting automations, data sync, API connections, Clay, n8n, CRM data pipelines